Supplementary Information

Bayesian Neural Surrogates Unify Uncertainty Propagation,
Sensitivity Attribution and Calibration for Coupled
Multi-Physics Reactor Analysis

补充信息

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1 Supplementary Figures / 补充图

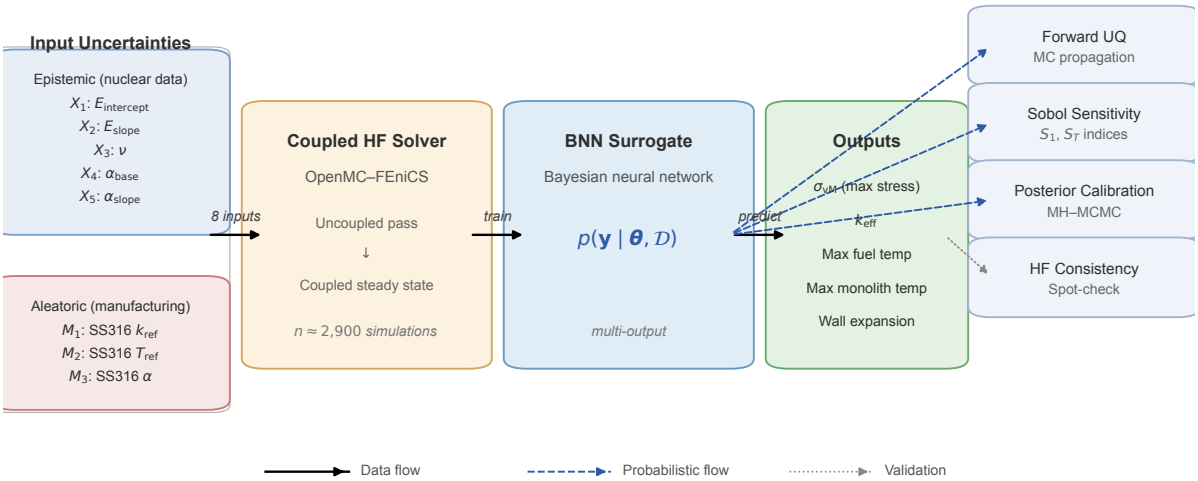

Figure 1: **Fig. S1 | BNN surrogate workflow.** Schematic of the end-to-end probabilistic analysis framework. The BNN surrogate is trained on coupled OpenMC–FEniCS data and produces a posterior predictive distribution. This single distribution serves as the shared probabilistic layer for forward uncertainty propagation, Sobol sensitivity attribution, and MCMC posterior calibration, without retraining between analyses.
图 S1 | BNN 代理工作流示意图。端到端概率分析框架。BNN 代理在耦合 OpenMC–FEniCS 数据上训练，产生后验预测分布，作为三类分析的共享概率层。

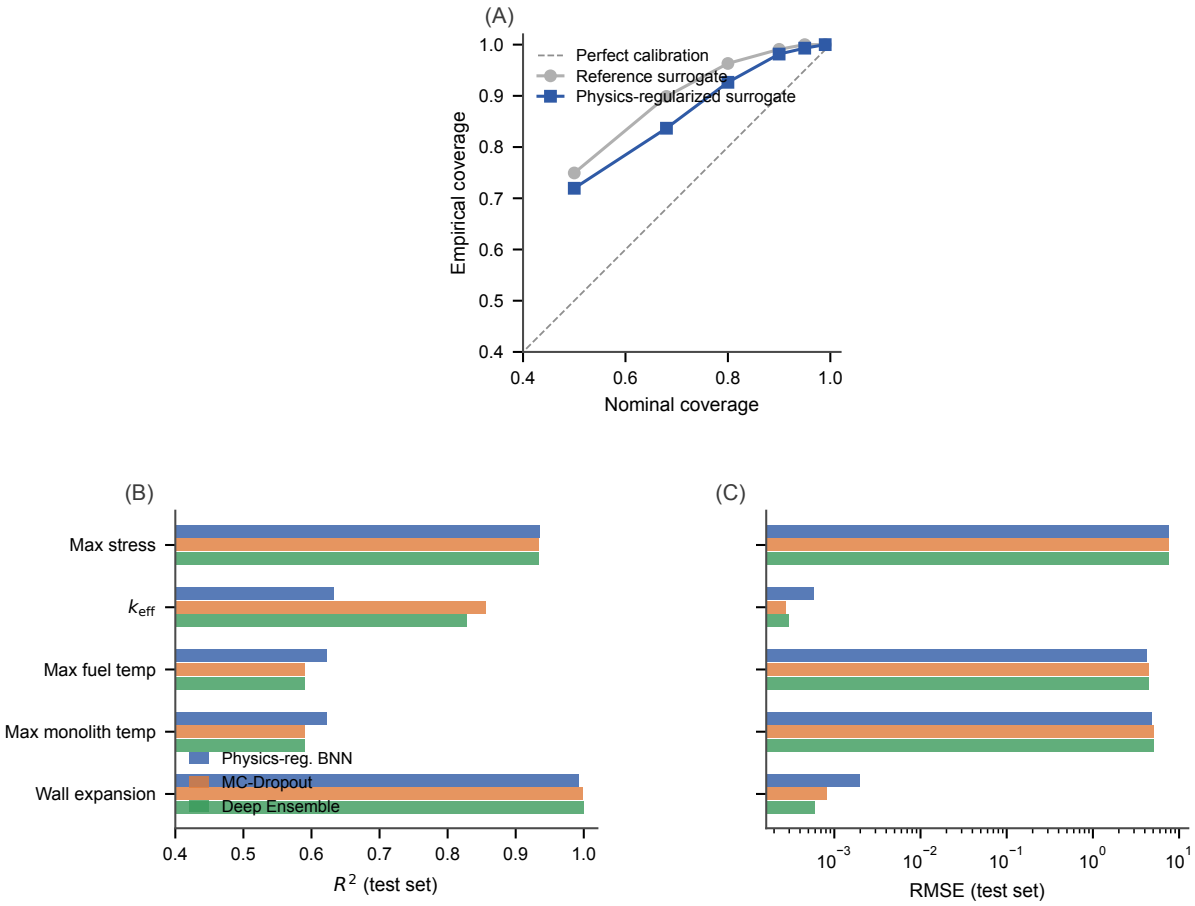

Figure 2: **Fig. S2 | Model validation: parity and residual analysis.** Predicted versus true values and residual distributions for primary outputs across all four BNN variants (435 test samples). Shaded regions show the 90% predictive interval.
图 S2 | 模型验证：一致性与残差分析。四种 BNN 变体在 435 个测试样本上的预测值与真实值及残差分布。

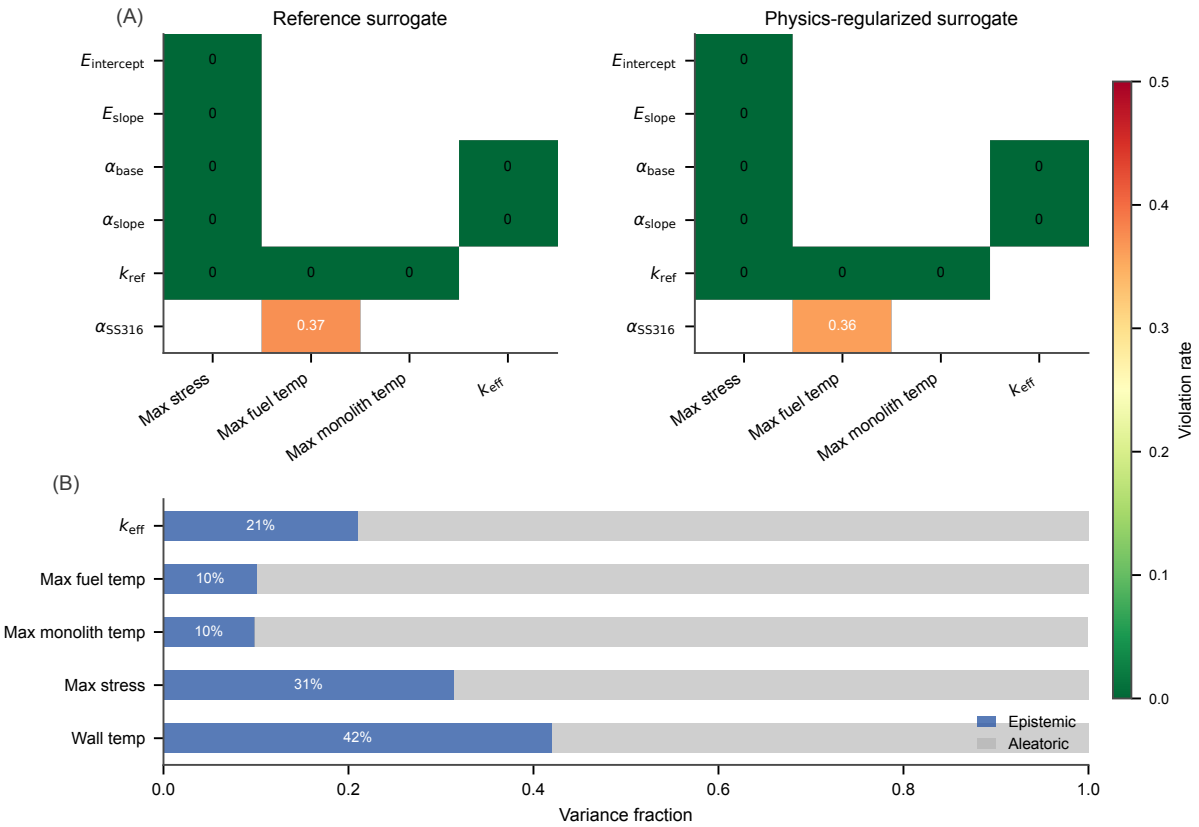

Figure 3: **Fig. S3 | Physics consistency and uncertainty decomposition.** (**A**) Monotonicity violation rates for input–output pairs across four BNN variants. (**B**) Epistemic vs. aleatoric variance decomposition. Aleatoric uncertainty dominates across all outputs (58–90% of total variance).
图 S3 | 物理一致性与不确定性分解。(A) 四种 BNN 变体的单调性违反率。(B) 认知与偶然方差分解。

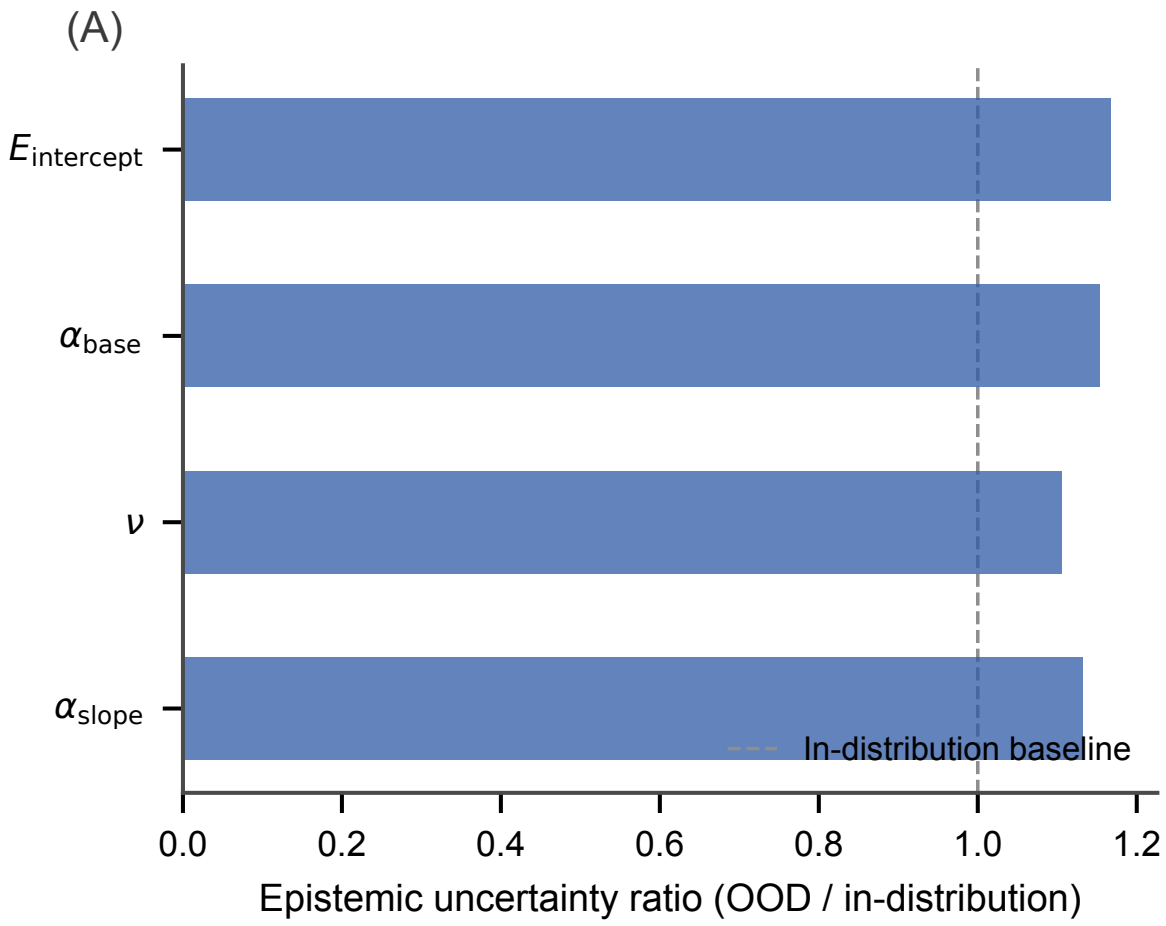

Figure 4: **Fig. S4 | Out-of-distribution (OOD) coverage and epistemic uncertainty.** 90% predictive interval coverage for in-distribution versus four feature-tail subsets. Stress retains coverage above 0.75 across all tails. k_{eff} coverage degrades substantially on the α_{base} tail (R^2 drops to approximately 0.29), indicating that the Sobol and posterior conclusions for k_{eff} are valid only within the training range of this dominant parameter. Elevated epistemic ratios on the α_{base} tail confirm that the BNN correctly signals increased model uncertainty outside the training domain.
图 S4 | 分布外覆盖率与认知不确定性。分布内与四个特征尾部子集的覆盖率对比。k_{eff} 在 α_{base} 尾部显著下降。

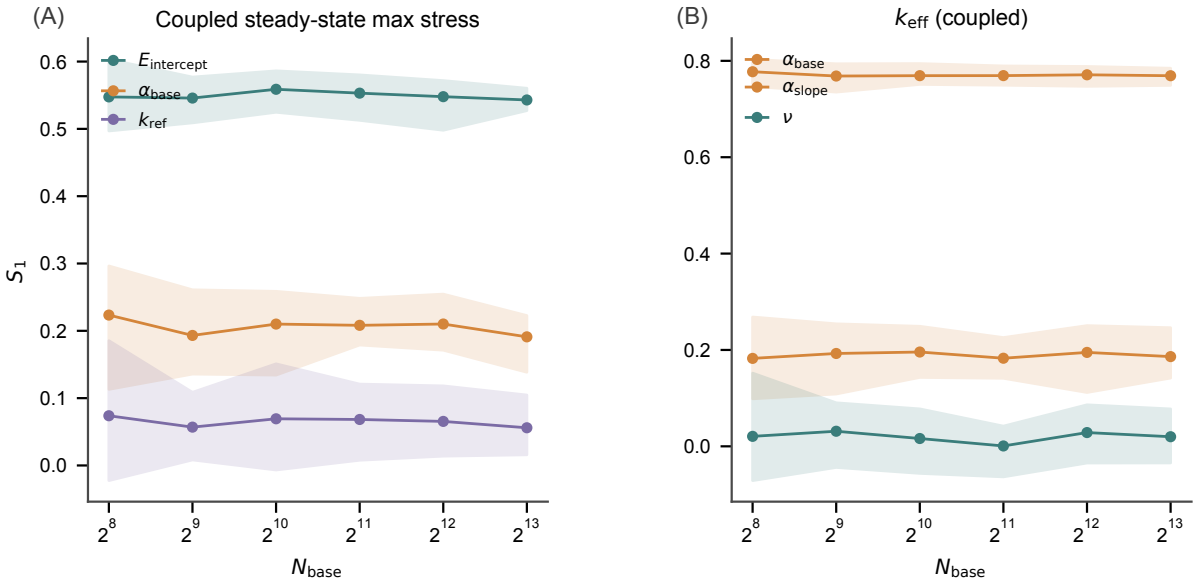

Figure 5: **Fig. S5 | Sobol index convergence with sample size.** First-order Sobol index S_1 as a function of sample size N_S (256 to 4096) for stress and k_{eff}, with 90% CI bands across 50 replications. Indices stabilise by $N_S = 2048$, confirming convergence at the chosen $N_S = 4096$.
图 S5 | Sobol 指标随样本量的收敛。应力和 k_{eff} 的一阶 Sobol 指标 S_1 随 N_S 变化，含 50 次重复的 90% CI 带。

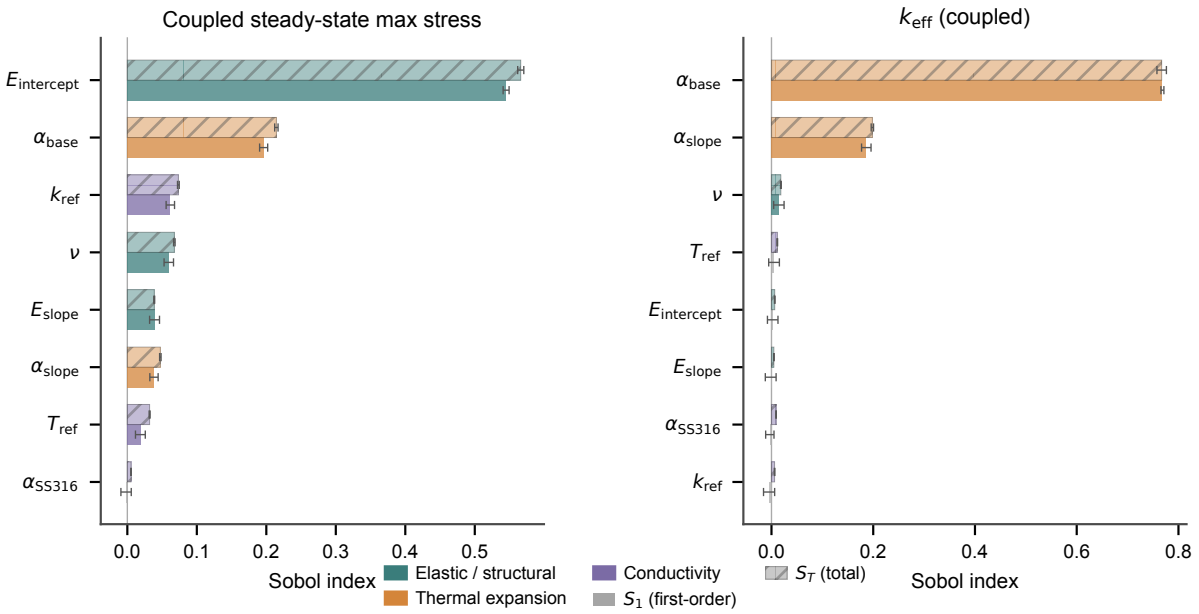

Figure 6: **Fig. S6 | Detailed Sobol sensitivity analysis.** First-order and total-order Sobol indices for all eight material parameters across all five primary outputs, showing the dominance of $E_{\text{intercept}}$ for stress and α_{base} for k_{eff}.
图 S6 | 详细 Sobol 敏感性分析。全部八个材料参数在五个主要输出上的一阶和全阶 Sobol 指标。

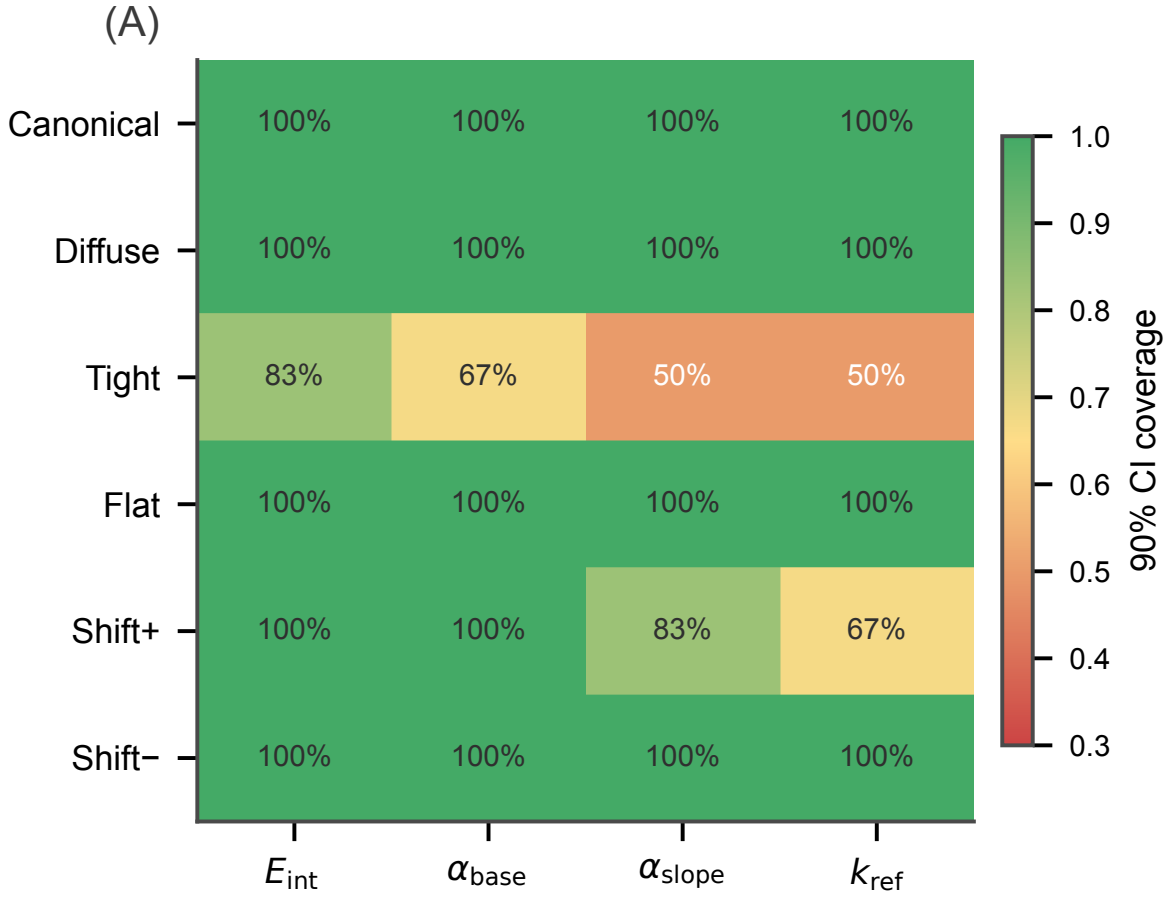

Figure 7: **Fig. S7 | Prior sensitivity analysis.** Posterior marginal distributions for the four calibrated parameters under six prior configurations: canonical, diffuse (2×), tight (0.5×), flat (uniform), shift$_+$ ($\mu + 0.5\sigma$), shift$_-$ ($\mu - 0.5\sigma$). The tight prior over-constrains the posterior, reducing coverage to 50–83% (Table S11).
图 S7 | 先验敏感性分析。六种先验配置下四个标定参数的后验边缘分布。紧凑先验过度约束后验。

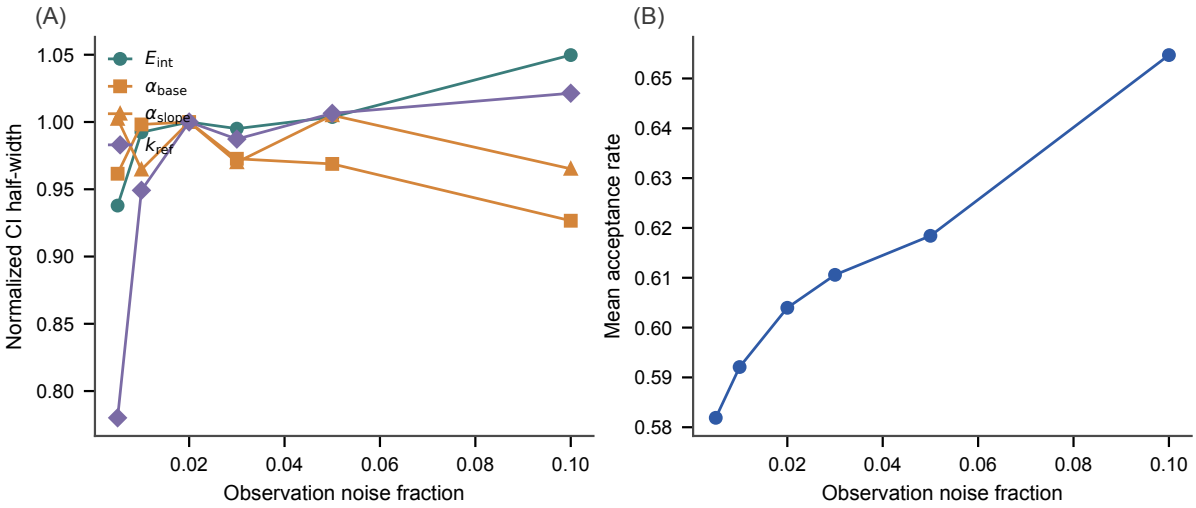

Figure 8: **Fig. S8 | Observation noise sensitivity.** Posterior width and acceptance rate as a function of assumed observation noise fraction (0.5%–10%). Posterior standard deviations increase monotonically; acceptance rates increase appropriately as the likelihood broadens (Table S12).
图 S8 | 观测噪声敏感性。后验宽度和接受率随假设观测噪声分数（0.5%–10%）的变化。

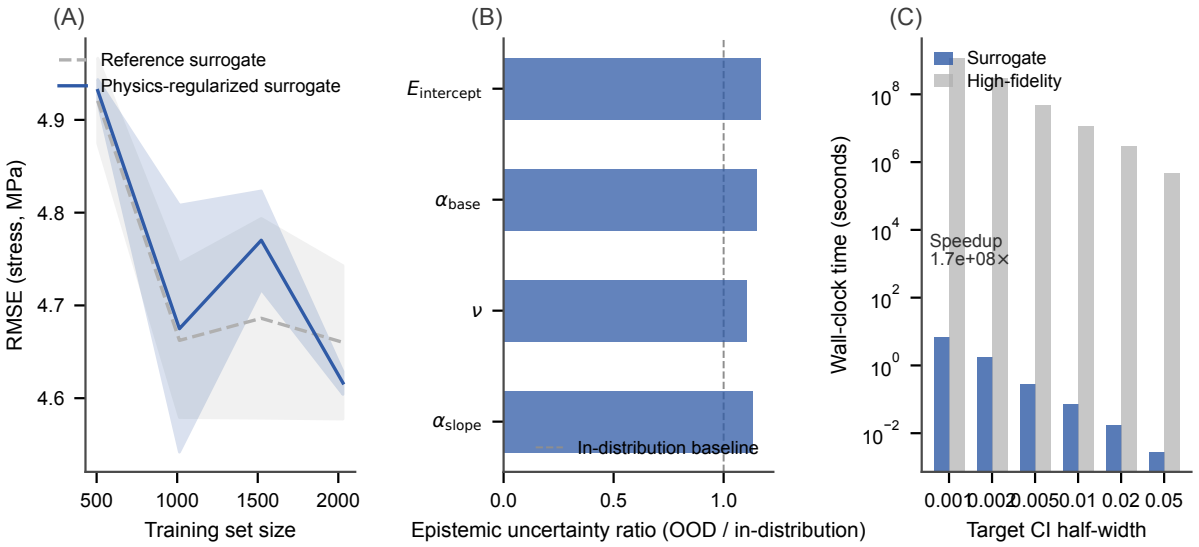

Figure 9: **Fig. S9 | Data efficiency and computational cost.** Test RMSE and R^2 as a function of training set size for the physics-regularized BNN. Near-saturated performance is achieved at 25% data utilisation (507 training samples; Table S13).
图 S9 | 数据效率与计算成本。物理正则化 BNN 的测试 RMSE 和 R^2 随训练集大小的变化。25% 数据即达到近饱和性能。

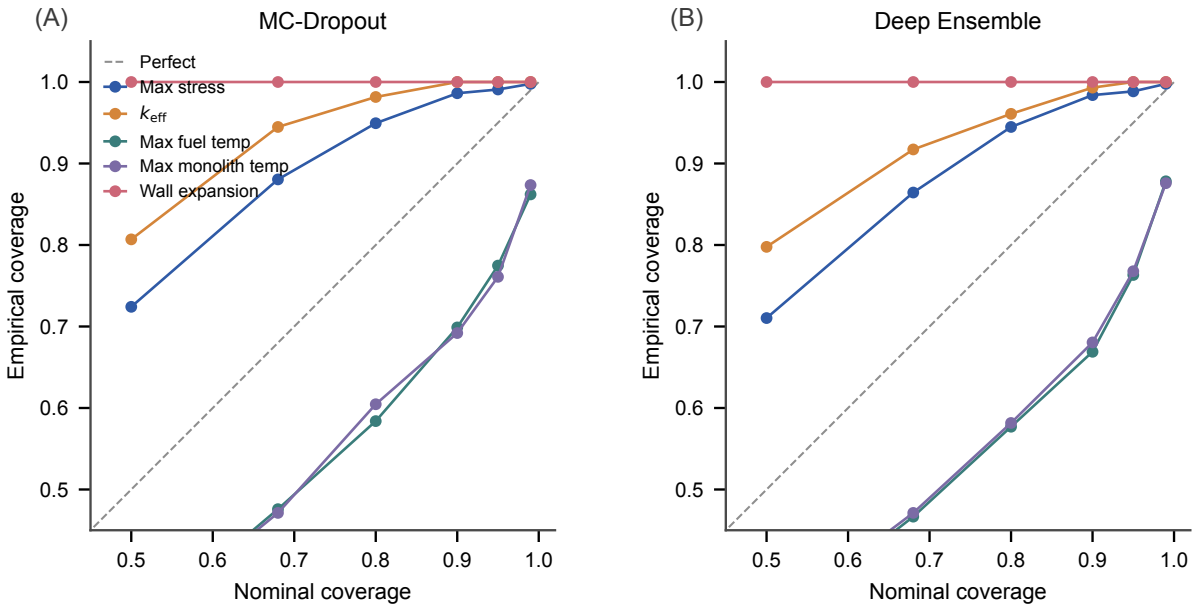

Figure 10: **Fig. S10 | External baseline calibration comparison.** Calibration curves comparing BNN, MC-Dropout and deep ensemble across primary outputs. The BNN achieves substantially better probabilistic calibration (ECE 0.027 vs. 0.18–0.19) on temperature outputs.
图 S10 | 外部基线校准对比。BNN、MC-Dropout 和深度集成的校准曲线。BNN 在温度输出上校准质量显著更优。

2 Supplementary Tables / 补充表

Table 1: **Table S1 | Material parameters: definitions, nominal values and priors.**
表 S1 | 材料参数：定义、标称值和先验。

Parameter	Symbol	Nominal value	Unit	Prior range ($\pm 10\%$)
Young's mod. intercept	$E_{\text{intercept}}$	2×10^{11}	Pa	$[1.8, 2.2] \times 10^{11}$
Young's mod. slope	E_{slope}	-7×10^{7}	Pa/K	$[-7.7, -6.3] \times 10^{7}$
Poisson's ratio	ν	0.31	—	[0.279, 0.341]
CTE base	α_{base}	1×10^{-5}	K^{-1}	$[0.9, 1.1] \times 10^{-5}$
CTE slope	α_{slope}	5×10^{-9}	K^{-2}	$[4.5, 5.5] \times 10^{-9}$
Ref. conductivity	k_{ref}	23.2	W/(m·K)	[20.88, 25.52]
Ref. temperature	T_{ref}	923.15	K	[830.84, 1015.47]
Cond. slope	k_{slope}	1/75	W/(m·K^2)	[0.012, 0.0147]

Table 2: **Table S2 | BNN hyperparameter search space and optimal values.**
表 S2 | BNN 超参数搜索空间与最优值。

Hyperparameter	Search range	Optimal (bnn-phy-mono)
Width	[64, 256]	254
Depth	[2, 8]	2
Learning rate	$[10^{-4}, 10^{-2}]$	9.81×10^{-4}
Weight decay	$[10^{-9}, 10^{-5}]$	8.09×10^{-8}
Batch size	{16, 32, 64}	32
Epochs	[200, 500]	399
Gradient clip	[0.1, 2.0]	0.674
w_{data}	[0.1, 5.0]	1.855
Prior σ	[0.1, 2.0]	0.478
KL weight	$[10^{-5}, 10^{-2}]$	1.93×10^{-4}
w_{mono}	[0.01, 1.0]	0.212
$\rho_{\text{abs,min}}$	[0.01, 0.5]	0.123
mono_topk	[10, 100]	53

Search: Optuna with 200 trials; objective: validation RMSE. Activation: SiLU throughout. Output heads: mean + log-variance (heteroscedastic).

Table 3: **Table S3 | BNN ablation study: four model variants on five primary outputs** (mean $\pm$ s.d., 5 seeds).
表 S3 | BNN 消融实验：四种模型变体在五个主要输出上的表现。

Model variant	Stress R^2	Stress PICP	k_{eff} R^2
Reference (ELBO only)	0.936 ± 0.007	0.995 ± 0.002	0.633 ± 0.260
Data-monotone	0.932 ± 0.005	0.998 ± 0.002	0.635 ± 0.260
Physics-regularized	0.935 ± 0.007	0.987 ± 0.007	0.633 ± 0.258
Data+inequality	0.924 ± 0.007	1.000 ± 0.001	0.617 ± 0.249

Table 4: **Table S4 | Complete Sobol indices (S_1 and S_T) for stress and k_{eff}** (physics-regularized BNN, $N_S = 4096$, 50 replications, 90% CI).
表 S4 | 应力和 k_{eff} 的完整 Sobol 指标。

Parameter	Stress S_1 [90% CI]	Stress S_T [90% CI]
E_{slope}	0.037 [−0.046, 0.100]	0.039 [0.038, 0.039]
$E_{\text{intercept}}$	0.548 [0.498, 0.572]	0.566 [0.562, 0.571]
ν	0.063 [0.034, 0.097]	0.068 [0.067, 0.069]
α_{base}	0.210 [0.171, 0.254]	0.214 [0.212, 0.217]
α_{slope}	0.038 [−0.013, 0.091]	0.047 [0.046, 0.049]
T_{ref}	0.019 [−0.025, 0.055]	0.032 [0.031, 0.033]
k_{ref}	0.065 [0.014, 0.118]	0.073 [0.072, 0.075]
SS316_α	−0.004 [−0.068, 0.034]	0.005 [0.005, 0.006]
Parameter	**k_{eff} S_1 [90% CI]**	**k_{eff} S_T [90% CI]**
E_{slope}	0.010 [−0.052, 0.062]	0.005 [0.004, 0.005]
$E_{\text{intercept}}$	0.008 [−0.065, 0.071]	0.006 [0.006, 0.007]
ν	0.029 [−0.035, 0.086]	0.018 [0.018, 0.019]
α_{base}	0.771 [0.747, 0.788]	0.767 [0.758, 0.776]
α_{slope}	0.195 [0.112, 0.250]	0.198 [0.196, 0.201]
T_{ref}	0.010 [−0.023, 0.054]	0.011 [0.011, 0.012]
k_{ref}	0.000 [−0.063, 0.048]	0.006 [0.006, 0.007]
SS316_α	0.003 [−0.059, 0.074]	0.009 [0.009, 0.009]

Table 5: **Table S5 | Posterior calibration: 18 benchmark case summary** (physics-regularized BNN).
表 S5 | 后验标定：18 个基准算例汇总。

Case	Category	True stress (MPa)	Accept rate	90%-CI coverage	Post-mean err.
0	low	110.6	0.619	3/4	8.3
1	low	118.7	0.612	4/4	24.6
2	low	110.7	0.620	2/4	9.0
3	low	103.7	0.618	4/4	8.8
4	low	114.5	0.621	4/4	3.1
5	low	102.0	0.616	3/4	23.4
6	near	127.6	0.609	4/4	20.3
7	near	127.9	0.586	4/4	26.6
8	near	127.0	0.611	4/4	5.9
9	near	123.5	0.613	4/4	3.5
10	near	128.9	0.609	3/4	4.5
11	near	129.3	0.610	4/4	3.8
12	high	169.5	0.592	4/4	0.2
13	high	176.0	0.591	4/4	12.1
14	high	193.4	0.598	3/4	13.7
15	high	163.4	0.591	4/4	3.6
16	high	142.8	0.582	4/4	10.5
17	high	197.6	0.583	4/4	5.4
Overall			0.58–0.62	66/72 = 0.917	MAE = 10.4

Table 6: **Table S6 | Computational cost breakdown.**
表 S6 | 计算成本分解。

Operation	Time	Platform
Coupled HF solve (1 case)	2266 s	EPYC 9654 / 96 cores
BNN single draw ($M = 50$)	1.29×10^{-2} s	GPU
BNN batched (batch=1024, per sample)	1.34×10^{-5} s	GPU
20,000-sample forward UQ	$\sim$0.3 s	GPU
Sobol (50 reps $\times$ 40,960 evals)	$\sim$minutes	GPU
18-case MCMC (8000 iter $\times$ 4 chains)	$\sim$minutes	GPU
Speedup (single draw vs. HF)	1.76×10^5	
Speedup (batched vs. HF)	1.69×10^8	

Table 7: **Table S7 | Physics consistency: monotonicity violation rates** (435 test samples, discrete $\pm\delta$ perturbation test).
表 S7 | 物理一致性：单调性违反率。

Physics pair (input $\rightarrow$ output, sign)	Ref. BNN	Data-mono	Phy-reg	Data+ineq
$E_{\text{intercept}} \rightarrow$ Max stress (+)	0%	0%	0%	0%
$\alpha_{\text{base}} \rightarrow$ Max stress (+)	0%	0%	0%	0%
$\alpha_{\text{base}} \rightarrow k_{\text{eff}}$ ($-$)	0%	0%	0%	0%
$k_{\text{ref}} \rightarrow$ Max stress ($-$)	0%	0%	0%	1.1%
SS316_$\alpha \rightarrow$ Fuel temp ($-$)	36.3%	39.5%	35.6%	53.6%

Inequality constraints (max $\geq$ average temp, stress ≥ 0): 0% violations for all four BNN variants.
不等式约束（最大 $\geq$ 平均温度，应力 ≥ 0）：四种 BNN 变体均为 0% 违反。

Table 8: **Table S8 | Epistemic vs. aleatoric uncertainty decomposition on test set.**
表 S8 | 测试集上的认知与偶然不确定性分解。

	Ref. BNN		Phy-reg BNN	
Output	Epi%	Ale%	Epi%	Ale%
k_{eff}	12.0	88.0	10.2	89.8
Max fuel temp	12.0	88.0	10.1	89.9
Max monolith temp	11.2	88.8	10.1	89.9
Max stress	27.7	72.3	29.6	70.4
Wall expansion	34.1	65.9	41.7	58.3

Table 9: **Table S9 | External baseline: full comparison of scoring rules.**
表 S9 | 外部基线：完整评分规则对比。

Model	Output	RMSE	R^2	CRPS	NLL	ECE
MC-Dropout	k_{eff}	0.0003	0.856	0.0002	-6.502	0.152
	Fuel temp	4.497	0.590	2.624	3.352	0.183
	Monolith T	5.061	0.590	2.951	3.472	0.182
	Max stress	7.659	0.934	4.519	3.592	0.118
	Wall exp.	0.0008	0.999	0.002	-3.831	0.197
Deep Ensemble	k_{eff}	0.0003	0.828	0.0002	-6.476	0.142
	Fuel temp	4.495	0.590	2.634	3.326	0.190
	Monolith T	5.061	0.590	2.962	3.439	0.185
	Max stress	7.644	0.934	4.502	3.586	0.112
	Wall exp.	0.0006	0.999	0.002	-3.841	0.197

Table 10: **Table S10 | Prior sensitivity** (6 configurations, physics-regularized BNN, case 06, near-threshold).
表 S10 | 先验敏感性（6 种配置，物理正则化 BNN，案例 06）。

Prior variant	Coverage (4 params)	Mean KL vs. canonical	Accept rate
Canonical	100%	0.00 (reference)	0.61
Diffuse ($2\times$)	100%	0.15–0.22	0.63
Tight ($0.5\times$)	50–83%	0.64–1.28	0.57
Flat (uniform)	100%	0.24–0.38	0.64
Shift ($+0.5\sigma$)	67–100%	0.06–0.36	0.60
Shift (-0.5σ)	83–100%	0.08–0.28	0.61

Table 11: **Table S11 | Noise sensitivity** (noise fraction 0.5%–10%, physics-regularized BNN, case 06).
表 S11 | 噪声敏感性（噪声分数 0.5%–10%）。

Noise fraction	$E_{\text{intercept}}$ post. std	Accept rate	Coverage
0.5%	1.26×10^{10}	0.58	100%
1.0%	1.28×10^{10}	0.59	100%
2.0% (default)	1.32×10^{10}	0.61	100%
5.0%	1.36×10^{10}	0.63	100%
10.0%	1.40×10^{10}	0.65	83%

Table 12: **Table S12 | Data efficiency** (2 seeds per fraction).
表 S12 | 数据效率（每个分数 2 个种子）。

Train fraction	N_{train}	RMSE (mean $\pm$ s.d.)	R^2 (mean $\pm$ s.d.)
25%	507	4.93 ± 0.04	0.768 ± 0.002
50%	1015	4.66 ± 0.11	0.782 ± 0.005
75%	1522	4.68 ± 0.01	0.784 ± 0.002
100%	2029	4.62 ± 0.08	0.783 ± 0.005

3 Supplementary Notes / 补充说明

3.1 Note A: BNN training details / BNN 训练细节

A.1. Evidence lower bound (ELBO) derivation / 证据下界推导

The BNN posterior over weights $\mathbf{w}$ is approximated by a mean-field variational distribution $q(\mathbf{w}) = \prod_i \mathcal{N}(w_i \mid \mu_i, \sigma_i^2)$. The variational parameters are optimised by minimising the negative evidence lower bound:

$$\mathcal{L}(\boldsymbol{\mu}, \boldsymbol{\sigma}) = \mathrm{KL}\big[q(\mathbf{w}) \,\|\, p(\mathbf{w})\big] - \mathbb{E}_{q(\mathbf{w})}\big[\log p(\mathcal{D} \mid \mathbf{w})\big] \tag{1}$$

where $p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I})$ is the isotropic Gaussian prior. The KL divergence admits a closed form; the data-fit term is approximated by a single Monte Carlo sample per mini-batch (Bayes-by-Backprop[10]).

BNN 的权重后验由均场变分分布近似。变分参数通过最小化负证据下界来优化。重参数化技巧使梯度计算成为可能：$w = \mu + \sigma \cdot \epsilon$，其中 $\epsilon \sim \mathcal{N}(0, \mathbf{I})$。方差参数 $\sigma = \log(1 + \exp(\rho))$ 以确保正性，$\rho_{\text{abs,min}}$ 钳制 $\rho \geq \rho_{\text{abs,min}}$ 以防止后验坍缩。

A.2. Heteroscedastic observation noise / 异方差观测噪声

The network outputs both a mean $\mu(\boldsymbol{\theta})$ and a log-variance $\log \sigma_{\text{obs}}^2(\boldsymbol{\theta})$ per output channel:

$$\log p(y_j \mid \boldsymbol{\theta}, \mathbf{w}) = -\tfrac{1}{2}\big[\log \sigma_{\text{obs},j}^2 + (y_j - \mu_j)^2 / \sigma_{\text{obs},j}^2\big] \tag{2}$$

该异方差公式允许 BNN 建模输入依赖的观测噪声，将其与认知不确定性分离。

A.3. KL annealing schedule / KL 退火策略

The KL weight β is annealed from 0 to its target value kl_weight $= 1.93 \times 10^{-4}$ over the first 100 epochs: $\beta(t) = \min(t/100, 1) \cdot$ kl_weight.

KL 权重通过线性策略在前 100 轮退火，防止 KL 项在数据拟合项找到合理权重空间区域之前主导早期训练。

A.4. Physics-prior monotonicity constraints / 物理先验单调性约束

The physics-regularized BNN adds an auxiliary loss:

$$\mathcal{L}_{\text{mono}} = \frac{1}{K} \sum_{k=1}^{K} \max\big(0,\ -\text{sign} \cdot \partial\mu / \partial\theta_k\big) \tag{3}$$

weighted by $w_{\text{mono}} = 0.212$. Sign constraints are derived from constitutive physics (e.g. stress increases with $E_{\text{intercept}}$).

物理正则化 BNN 添加辅助损失项，惩罚违反从本构律推导的已知单调性关系。

A.5. Gradient clipping and initialisation / 梯度裁剪与初始化

Gradient norms are clipped at 0.674. Weight means μ are initialised with Xavier uniform; ρ values are initialised uniformly in $[-5, -4]$.

梯度范数按参数组裁剪至 0.674。权重均值以 Xavier 均匀初始化；ρ 值在 $[-5,-4]$ 上均匀初始化。

3.2 Note B: Sobol sensitivity methodology / Sobol 敏感性方法

B.1. Sobol variance decomposition / 方差分解

For a square-integrable function $f(\theta_1,\ldots,\theta_d)$ with independent inputs:

$$V=\sum_i V_i+\sum_{i<j} V_{ij}+\cdots+V_{1,\ldots,d} \tag{4}$$

The first-order Sobol index is $S_i=V_i/V$; the total-order index is $S_{T,i}=1-V_{\sim i}/V$.

一阶 Sobol 指标 $S_i=V_i/V$；全阶指标 $S_{T,i}=1-V_{\sim i}/V$。

B.2. Saltelli cross-matrix scheme / 交叉矩阵方案

Two independent $N\times d$ matrices $\mathbf{A}$ and $\mathbf{B}$ are drawn from the joint prior. For each input i, a cross-matrix $\mathbf{A}_{\mathbf{B}}^{(i)}$ is constructed. The model is evaluated at all $N(d+2)$ points.

从联合先验中抽取两个独立矩阵，对每个输入构造交叉矩阵，在所有 $N(d+2)$ 个点上评估模型。

B.3. Jansen estimator / Jansen 估计量

$$S_{1,i}=1-\frac{1}{2N}\sum_{j=1}^{N}\big[f(\mathbf{A}_{\mathbf{B}}^{(i)})_j-f(\mathbf{B})_j\big]^2/V \tag{5}$$

$$S_{T,i}=\frac{1}{2N}\sum_{j=1}^{N}\big[f(\mathbf{A})_j-f(\mathbf{A}_{\mathbf{B}}^{(i)})_j\big]^2/V \tag{6}$$

B.4. Replication-based confidence intervals / 基于重复的置信区间

The entire Sobol analysis is independently repeated $R=50$ times. The 90% CI is constructed from the 5th and 95th percentiles. The BNN posterior predictive mean ($M=30$ weight samples) serves as f.

完整 Sobol 分析独立重复 $R=50$ 次。BNN 后验预测均值（$M=30$）充当 f。

3.3 Note C: MCMC posterior calibration details / MCMC 后验标定细节

C.1. Likelihood formulation / 似然公式

$$\log p(\mathbf{y}_{\text{obs}}\mid\boldsymbol{\theta})=\sum_{j=1}^{J}\log\mathcal{N}\big(y_{\text{obs},j}\mid\mu_j(\boldsymbol{\theta}),\ \sigma_{\text{obs},j}^2+\sigma_{\text{BNN},j}^2(\boldsymbol{\theta})\big) \tag{7}$$

where μ_j is the BNN predictive mean ($M=20$ samples) and $\sigma_{\text{obs}}=2\%$ of the true value. $J=5$ primary outputs.

其中 μ_j 为 BNN 预测均值，σ_{obs} 为真实值的 2%。

C.2. Proposal distribution / 提议分布

Symmetric Gaussian random-walk: $\boldsymbol{\theta}^* = \boldsymbol{\theta}_{\text{current}} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{proposal}})$.

对称高斯随机游走提议，尺度通过 1000 次迭代的短试探运行校准。

C.3. Chain configuration / 链配置

Per benchmark case: 4 independent chains, $N_{\text{total}} = 8000$ iterations, burn-in 2000, thinning 5. Effective posterior: 1200 samples per chain, 4800 total. Calibrated parameters: $E_{\text{intercept}}$, α_{base}, α_{slope}, k_{ref}.

每个基准算例：4 条独立链，8000 次迭代，预热 2000 次，稀疏化步长 5。

C.4. Convergence diagnostics / 收敛诊断

Rank-normalised split-$\hat{R}$: $\hat{R}_{\text{max}} = 1.017$ (all 72 parameter–case combinations). $\text{ESS}_{\text{min}} = 352$. Acceptance rates: 0.58–0.62 (mean 0.605).

秩归一化 split-$\hat{R}$：$\hat{R}_{\text{max}} = 1.017$，$\text{ESS}_{\text{min}} = 352$，接受率 0.58–0.62。

C.5. Benchmark case selection / 基准算例选取

18 cases from the 435-sample test split: 6 low (102–119 MPa), 6 near (123–129 MPa), 6 high (143–198 MPa). Threshold: 131 MPa. Observations: true HF outputs $+2\%$ multiplicative Gaussian noise.

18 个基准算例取自测试集，覆盖低、阈值附近和高应力区间。

3.4 Note D: Forward UQ extensions / 前向 UQ 扩展

D.1. Predictive mean vs. full draw / 预测均值与完整抽样

The main-text forward UQ propagates through the BNN posterior predictive mean $y = \mathbb{E}_{\mathbf{w}}[\mu_{\mathbf{w}}(\boldsymbol{\theta})]$. As a consistency check, full predictive draws (sampling both weight uncertainty and observation noise) produce wider distributions but consistent means and percentile rankings.

正文前向 UQ 通过 BNN 后验预测均值传播。完整预测抽样分布更宽，但均值和百分位排名一致。

D.2. Latin hypercube sampling / 拉丁超立方采样

LHS ensures more uniform coverage of the input space than simple random sampling. $N = 20{,}000$ samples, $d = 8$ inputs.

LHS 确保比简单随机采样更均匀的输入空间覆盖。

3.5 Note E: Coupled simulation and material model / 耦合仿真与材料模型

E.1. Material constitutive models / 材料本构模型

SS316 monolith parameters grouped by physical function:

- Structural mechanics: $E(T) = E_{\text{slope}} \cdot T + E_{\text{intercept}}$; ν (Poisson's ratio). Stress: $\sigma = \mathbf{C}(E, \nu) \cdot \boldsymbol{\epsilon}$.

- Thermal deformation: $\alpha(T) = \alpha_{\text{base}} + \alpha_{\text{slope}} \cdot T$. Dual role: thermal strain $\epsilon_{\text{th}} = \alpha \cdot \Delta T$ and geometric expansion.
- Thermal transport: $k(T) = k_{\text{ref}} + k_{\text{slope}} \cdot (T - T_{\text{ref}})$.

SS316 一体化结构参数按物理功能分组：结构力学、热变形、热输运。

E.2. Picard iteration / Picard 迭代

OpenMC–FEniCS coupling: (1) initialise; (2) OpenMC computes k_{eff} and fission power; (3) FEniCS solves temperature, displacement, stress; (4) feedback updated geometry; (5) repeat until $\Delta T_{\text{max}} < 1\,\text{K}$. Under-relaxation factor: 0.1.

OpenMC–FEniCS 耦合迭代，收敛准则：$\Delta T_{\text{max}} < 1\,\text{K}$，欠松弛因子 0.1。

E.3. Output definitions / 输出定义

Each HF evaluation produces 15 scalar outputs: 7 decoupled (before coupling convergence) and 8 coupled (after full convergence, including k_{eff}). The posterior likelihood uses 5 primary outputs: k_{eff}, peak fuel temperature, peak monolith temperature, peak global stress, and wall expansion.

每次高保真评估产生 15 个标量输出：7 个解耦和 8 个耦合。后验似然使用 5 个主要输出。

E.4. Dataset generation / 数据集生成

$n = 2900$ samples by LHS; split (seed $= 2026$): training 2029, validation 436, test 435.

通过拉丁超立方采样生成 2900 个样本；固定划分复用以确保可重复性。

3.6 Note F: High-fidelity coupled simulation framework / 高保真耦合仿真框架

This note provides the governing equations of the four solver modules that generate the coupled training data consumed by the BNN surrogate. The framework couples OpenMC (neutron transport), FEniCS (thermo-elasticity), a one-dimensional heat-pipe thermal–hydraulic model and a fuel-temperature module via Picard fixed-point iteration. Details follow the implementation described in Refs. [9, 11].

本节给出生成 BNN 代理训练数据的四个求解器模块的控制方程。该框架通过 Picard 不动点迭代耦合 OpenMC（中子输运）、FEniCS（热弹性）、一维热管热工水力模型和燃料温度模块。实现细节参见文献 [9, 11]。

F.1. Neutron transport — OpenMC / 中子输运 — OpenMC

OpenMC [11] is an open-source Monte Carlo particle transport code supporting fixed-source, k-eigenvalue and subcritical multiplication calculations. In the coupled framework it computes k_{eff}, the fission power distribution and auxiliary neutronic tallies.

OpenMC [11] 是一款开源蒙特卡罗粒子输运程序，支持固定源、k-特征值及次临界倍增计算。在耦合框架中它负责计算 k_{eff}、裂变功率分布和辅助中子学参数。

Integral transport equation. The steady-state, continuous-energy neutron transport equation in integral form reads

$$\Phi(\mathbf{r}, \mathbf{\Omega}, E) = \int_0^\infty e^{-\tau(\mathbf{r}, \mathbf{r}-s\mathbf{\Omega}, E)}\, Q(\mathbf{r} - s\mathbf{\Omega}, \mathbf{\Omega}, E)\, \mathrm{d}s \tag{8}$$

where Φ is the angular neutron flux at position $\mathbf{r}$, direction $\mathbf{\Omega}$ and energy E; Q is the emission density (scattering + fission source); and τ is the optical thickness:

$$\tau(\mathbf{r}, \mathbf{r}', E) = \int_0^{|\mathbf{r}-\mathbf{r}'|} \Sigma_t(\mathbf{r} - s\mathbf{\Omega}, E)\, \mathrm{d}s \tag{9}$$

with Σ_t the macroscopic total cross-section.

稳态连续能量中子输运方程的积分形式如式 (8)。其中 Φ 为角中子通量，Q 为发射密度（散射源 + 裂变源），τ 为光学厚度，Σ_t 为宏观总截面。

Emission density and Fredholm formulation. Substituting the scattering and fission kernels into the emission density yields the second-kind Fredholm integral equation:

$$Q(\mathbf{r}, \mathbf{\Omega}, E) = \int K(\mathbf{r}', \mathbf{\Omega}', E' \to \mathbf{r}, \mathbf{\Omega}, E)\, Q(\mathbf{r}', \mathbf{\Omega}', E')\, \mathrm{d}\mathbf{r}'\, \mathrm{d}\mathbf{\Omega}'\, \mathrm{d}E' \tag{10}$$

where the integral kernel K is the product of a transport kernel (free-flight attenuation) and a collision kernel (scattering + fission). Writing this in operator form $Q = \mathcal{K}Q$ and iterating:

$$Q^{(n+1)} = \mathcal{K}\, Q^{(n)}, \qquad Q = \lim_{n\to\infty} Q^{(n)} = \sum_{k=0}^{\infty} \mathcal{K}^k\, Q^{(0)} \tag{11}$$

which is the Neumann series solution. The converged Q is substituted back into Eq. (8) to recover the full neutron flux distribution.

将散射核和裂变核代入发射密度方程可得到第二类 Fredholm 积分方程。以算子形式写作 $Q = \mathcal{K}Q$ 并迭代，即得 Neumann 级数解。收敛的 Q 代回式 (8) 即可得到完整的中子通量分布。

Monte Carlo k-eigenvalue calculation. In practice, the integral equation is solved stochastically: a population of N_p neutron histories is tracked through successive fission generations. Each generation produces an estimate of

$$k_{\text{eff}} = \frac{\text{neutron production in generation } n+1}{\text{neutron production in generation } n} \tag{12}$$

and the converged fission source gives the spatial power distribution. The statistical uncertainty on k_{eff} is kept below 10 pcm (1σ) and the maximum power-distribution standard deviation below 0.19 % by using a sufficiently large number of particle histories per batch.

蒙特卡罗 k-特征值计算通过逐代追踪 N_p 个中子历史来求解积分方程。每代产生一个 k_{eff} 估计值，收敛的裂变源给出空间功率分布。通过足够多的粒子历史数使 k_{eff} 统计不确定度低于 10 pcm，功率分布最大标准差低于 0.19 %。

Geometry and temperature feedback. Thanks to the constructive solid geometry representation in OpenMC, the framework updates fuel-rod and heat-pipe coordinates (reflecting thermal expansion), ma-

terial temperatures (Doppler broadening) and material densities (thermal expansion) between coupling iterations.

得益于 OpenMC 的构造实体几何表示，框架可在耦合迭代间更新燃料棒和热管坐标（反映热膨胀）、材料温度（多普勒展宽）和材料密度（热膨胀）。

F.2. Thermo-elastic analysis — FEniCS / 热弹性分析 — FEniCS

The thermo-elastic module is built on the open-source finite-element platform FEniCS [12]. It solves two coupled problems on the monolith mesh: steady-state heat conduction and linear thermoelasticity.

热弹性模块基于开源有限元平台 FEniCS [12] 搭建，在基体网格上求解两个耦合问题：稳态导热和线性热弹性。

Heat conduction. The steady-state heat equation on the monolith domain Ω is

$$-\nabla \cdot \big[k(T)\,\nabla T\big] = \dot{q} \qquad \text{in } \Omega \tag{13}$$

where $k(T)$ is the temperature-dependent thermal conductivity of SS316 (see Note E) and $\dot{q}$ is the volumetric heat generation rate from fission. Boundary conditions are: isothermal walls at heat-pipe surfaces (temperatures from the heat-pipe module) and prescribed heat fluxes at fuel-rod surfaces (from the power distribution).

基体域 Ω 上的稳态导热方程如式 (13)，其中 $k(T)$ 为 SS316 温度依赖热导率，$\dot{q}$ 为裂变体积释热率。边界条件为：热管表面等温壁（温度由热管模块给出）和燃料棒表面热通量（由功率分布给出）。

Linear thermoelasticity. For an isotropic material, the three-dimensional linear elastic constitutive relation (generalised Hooke's law) is

$$\boldsymbol{\sigma} = \lambda\,\mathrm{tr}(\boldsymbol{\varepsilon})\,\mathbf{I} + 2\mu\,\boldsymbol{\varepsilon} \tag{14}$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor, $\boldsymbol{\varepsilon}$ the strain tensor, $\mathbf{I}$ the identity tensor, and λ, μ are the Lamé constants determined by the temperature-dependent Young's modulus $E(T)$ and Poisson's ratio ν:

$$\lambda = \frac{E\,\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)} \tag{15}$$

对各向同性材料，三维线弹性本构关系（广义胡克定律）如式 (14)。其中 λ, μ 为拉梅常数，由温度依赖的杨氏模量 $E(T)$ 和泊松比 ν 确定，见式 (15)。

Thermal strain (Duhamel–Neumann law). Including thermal deformation, the constitutive equation for an isotropic solid becomes

$$\boldsymbol{\sigma} = \lambda\,\mathrm{tr}(\boldsymbol{\varepsilon})\,\mathbf{I} + 2\mu\,\boldsymbol{\varepsilon} - (3\lambda + 2\mu)\,\alpha(T)\,\Delta T\,\mathbf{I} \tag{16}$$

where $\alpha(T) = \alpha_{\text{base}} + \alpha_{\text{slope}} \cdot T$ is the temperature-dependent coefficient of thermal expansion (CTE) and $\Delta T = T - T_0$ is the temperature rise from the stress-free reference state. The thermal strain tensor is $\boldsymbol{\varepsilon}_{\text{th}} = \alpha\,\Delta T\,\mathbf{I}$.

考虑热变形后，各向同性固体的本构方程变为 Duhamel–Neumann 定律，式 (16)。其中 $\alpha(T)$ 为温度依赖的线性热膨胀系数，ΔT 为相对无应力参考状态的温升，热应变张量 $\boldsymbol{\varepsilon}_{\text{th}} = \alpha\,\Delta T\,\mathbf{I}$。

Equilibrium equation. The static momentum balance (in the absence of body forces other than thermal loading) is

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \qquad \text{in } \Omega \tag{17}$$

which, combined with Eqs. (16) and the strain–displacement relation $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T)$, yields the displacement field $\mathbf{u}$ and the von Mises stress σ_{vM} used as the primary stress output.

静力平衡方程与式 (16) 及应变-位移关系联立，求解位移场 $\mathbf{u}$ 和 von Mises 应力 σ_{vM}（作为主要应力输出）。

F.3. Heat-pipe thermal–hydraulic model / 热管热工水力模型

Each sodium heat pipe is modelled with one-dimensional transient conservation equations for the vapour core and the porous wick. The model captures evaporation, vapour transport, condensation and capillary return, producing the steady-state wall temperature for a given input power [9].

每根钠热管采用一维瞬态守恒方程（蒸汽腔和多孔吸液芯）建模，捕获蒸发、蒸汽输运、冷凝和毛细回流，对给定输入功率产生稳态壁温 [9]。

Vapour continuity.

$$\frac{\partial \rho_v}{\partial t} + \frac{\partial(\rho_v\, u_v)}{\partial z} = \frac{\dot{m}_v}{D_v} \tag{18}$$

where ρ_v, u_v are the vapour density and axial velocity, z the axial coordinate, $\dot{m}_v$ the mass source from evaporation and D_v the vapour-core diameter.

蒸汽连续性方程：ρ_v、u_v 为蒸汽密度和轴向速度，$\dot{m}_v$ 为蒸发产生的质量源项，D_v 为蒸汽腔直径。

Liquid continuity in the wick.

$$\frac{\partial(\phi\, S\, \rho_l)}{\partial t} + \frac{\partial(\phi\, S\, \rho_l\, u_l)}{\partial z} = \frac{\dot{m}_l}{D_w} \tag{19}$$

where ϕ is the wick porosity, S the liquid fraction, ρ_l and u_l the liquid density and velocity, $\dot{m}_l$ the condensation mass source.

吸液芯中液体连续性方程：ϕ 为孔隙率，S 为液相份额，$\dot{m}_l$ 为冷凝质量源项。

Dusty-gas vapour transport. To handle the transition between free-molecular and continuum flow regimes inside the vapour core, a second-order dusty-gas model [13] is employed:

$$\frac{\partial(\rho_v\, u_v)}{\partial z} + \frac{\partial}{\partial z}\Big(\rho_v\, u_v^2 + \rho_v\, R_g\, T\Big) = -F_{\text{wall}} \tag{20}$$

where R_g is the specific gas constant and F_{wall} represents the axial flow-diffusion coefficient including wall–molecule collisions.

为处理蒸汽腔内自由分子流到连续流的过渡，采用了含尘气体模型的二阶近似 [13]。

Darcy's law for wick flow. Liquid flow through the porous wick is modelled by Darcy's law:

$$u_l = -\frac{\kappa}{\mu_l}\frac{\partial P_l}{\partial z} \tag{21}$$

where κ is the permeability of the wire-mesh wick:

$$\kappa = \frac{d_w^2\,\phi^3}{122\,(1-\phi)^2} \tag{22}$$

with d_w the wire diameter. The capillary pressure is $P_c = 2\gamma\cos\theta/r_p$, where γ is the surface tension and r_p the effective pore radius.

液体在多孔吸液芯中的流动由达西定律描述。渗透率 κ 由丝径 d_w 和孔隙率 ϕ 确定。毛细压力 $P_c = 2\gamma\cos\theta/r_p$。

Power–temperature database. Because the full transient heat-pipe simulation is expensive, a pre-computed database mapping input power to steady-state wall temperature is used during coupling iterations (420 data points, 1000–8000 W in 100 W steps). Linear extrapolation is applied for out-of-range powers.

由于完整瞬态热管模拟计算代价高，耦合迭代中使用预计算的输入功率– 稳态壁温数据库（420 个数据点，1000–8000 W，步长 100 W）。超出范围的功率采用线性外推。

F.4. Fuel temperature model / 燃料温度模型

The fuel temperature module computes fuel centreline and average temperatures from the monolith temperature field and the fission power distribution.

燃料温度模块根据基体温度场和裂变功率分布计算燃料中心线温度和平均温度。

Gas-gap conductance. The fuel–monolith gap is treated as a thin helium layer. The helium thermal conductivity is [14]:

$$k_{\mathrm{He}}(T) = A\,T^B \tag{23}$$

where A and B are empirical constants ($B = 0.645$ for helium). The temperature drop across the gap is

$$T_{\text{fuel,outer}} - T_{\text{gap,outer}} = \frac{q'}{2\pi\,k_{\mathrm{He}}}\,\ln\!\left(\frac{d_{\text{gap}}}{d_{\text{fuel}}}\right) \tag{24}$$

where q' is the linear power density, d_{gap} the gap outer diameter and d_{fuel} the fuel diameter.

燃料–基体间隙视为薄层氦气导热。间隙温降由线功率密度 q' 和氦气热导率决定。

Fuel centreline temperature. For a cylindrical fuel rod with internal heat generation and constant conductivity k_f, the centreline-to-surface temperature difference is

$$T_{\text{centre}} - T_{\text{fuel,outer}} = \frac{q'}{4\pi\,k_f} = \frac{\bar{q}\,r_f^2}{4\,k_f} \tag{25}$$

where $\bar{\dot{q}}$ is the average volumetric heat generation rate and r_f the fuel radius. The fuel average temperature is obtained by radial integration:

$$T_{\text{avg}} - T_{\text{fuel,outer}} = \frac{\bar{\dot{q}}\, r_f^2}{8\, k_f} \tag{26}$$

对具有内热源的圆柱形燃料棒，中心线与表面温差由线功率密度和燃料热导率决定。平均温度由径向积分得到。

F.5. Picard coupling iteration / Picard 耦合迭代

The four modules are coupled via Picard (fixed-point) iteration, which sequentially decouples the neutronics, thermal, mechanical and heat-pipe sub-problems. The iteration proceeds as follows:

四个模块通过 Picard（不动点）迭代耦合，依次解耦中子学、热学、力学和热管子问题。迭代步骤如下：

1. **Neutronics:** OpenMC computes k_{eff}, fission power distribution and key neutronic parameters.
2. **Power mapping:** per-fuel heating tallies are normalised to the rated reactor power to obtain volumetric heat generation rates and surface heat fluxes.
3. **Heat-pipe wall temperature:** the heat-pipe module (or database) converts each pipe's input power to a steady-state wall temperature.
4. **Thermo-elasticity:** FEniCS solves the heat equation (Eq. 13) and the thermoelastic equations (Eqs. 16) on the monolith mesh, yielding temperature, displacement and stress fields. Heat-pipe input powers are updated from the new temperature field.
5. **Inner convergence:** steps 3–4 are repeated until max $|\Delta T_{\text{HP}}| < 1\,\text{K}$ between successive iterations, with an under-relaxation factor of 0.1 applied to the heat-pipe wall temperatures.
6. **Fuel temperatures:** the fuel-temperature module computes centreline and average temperatures from the converged monolith temperature field.
7. **Geometry update:** displacement-driven coordinate updates for fuel rods and heat pipes are fed back to the OpenMC geometry.
8. **Outer convergence:** material temperatures, densities and coordinates are updated in OpenMC; neutron transport is re-run. The outer loop continues until physical quantities converge or a maximum iteration count is reached.

1. **中子学：** OpenMC 计算 k_{eff}、裂变功率分布和关键中子学参数。
2. **功率映射：**将各燃料单元加热计数归一化至额定反应堆功率，得到体积释热率和表面热通量。
3. **热管壁温：**热管模块（或数据库）将每根管的输入功率转换为稳态壁温。
4. **热弹性：** FEniCS 在基体网格上求解导热方程和热弹性方程，得到温度、位移和应力场；根据新温度场更新热管输入功率。

5. **内层收敛：**重复步骤 3–4，直到连续两步热管壁温最大差值 $< 1\,\mathrm{K}$（欠松弛因子 0.1）。
6. **燃料温度：**燃料温度模块根据收敛的基体温度场计算中心线和平均温度。
7. **几何更新：**基于位移的燃料棒和热管坐标更新反馈至 OpenMC 几何。
8. **外层收敛：**更新 OpenMC 中的材料温度、密度和坐标，重新运行中子输运，直到物理量收敛或达到最大迭代次数。

The neutronics statistical uncertainty is controlled to $< 10\,\mathrm{pcm}$ (1σ) on k_{eff} and $< 0.19\%$ maximum standard deviation on the power distribution. A single coupled solve requires approximately 2266 s on 96 cores (AMD EPYC 9654), with neutronics accounting for ∼67% of the total time (see Table S6).

中子学统计不确定度控制在 $k_{\mathrm{eff}} < 10\,\mathrm{pcm}$ (1σ)、功率分布最大标准差 $< 0.19\%$。单次耦合求解在 96 核（AMD EPYC 9654）上约需 2266 s，其中中子学约占总时间的 67%（见表 S6）。

F.6. MegaPower reactor model / MegaPower 反应堆模型

The coupled framework is applied to the MegaPower reactor, a $5\,\mathrm{MW_{th}}$ heat-pipe-cooled microreactor designed by Los Alamos National Laboratory [15]. Key design parameters:

耦合框架应用于 MegaPower 反应堆，该堆由洛斯阿拉莫斯国家实验室设计，热功率 $5\,\mathrm{MW_{th}}$ [15]。关键设计参数：

- **Monolith:** SS316 stainless steel; hexagonal lattice containing fuel rods and heat pipes. 基体为 SS316 不锈钢，六角栅格排列燃料棒和热管。
- **Fuel:** 19.75 % enriched UO_2 rods ($r_f = 0.71\,\mathrm{cm}$), 10 rows from inner core to outer core. 燃料为 19.75% 富集 UO_2 棒，10 排从内堆芯到外堆芯。
- **Heat pipes:** sodium heat pipes ($r_{\mathrm{HP}} = 0.98\,\mathrm{cm}$), evaporator length 1500 mm, adiabatic section 300 mm, condenser 2200 mm. 钠热管，蒸发段 1500 mm，绝热段 300 mm，冷凝段 2200 mm。
- **Reflector:** Al_2O_3 radial reflector with six rotating B_4C control drums for reactivity control. 径向 Al_2O_3 反射层含六个旋转 B_4C 控制鼓调节反应性。
- **Symmetry:** a 1/6-core sector is modelled with reflective boundary conditions to reduce computational cost. 利用 1/6 堆芯对称性建模以降低计算成本。

The FEniCS mesh resolves the monolith cross-section with fuel-rod holes (small) and heat-pipe holes (large). Plane-strain conditions are assumed for the two-dimensional thermo-elastic calculation; a simplified three-dimensional mesh is used for validation.

FEniCS 网格分辨基体横截面的燃料棒孔（小）和热管孔（大）。二维热弹性计算假设平面应变条件；简化三维网格用于验证。

Supplementary References / 补充参考文献

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